**6.5 Schematic diagram showing one aspect of the Teller–Ulam/Sakharov–Zel'-dovich idea for the design of a hydrogen bomb: A fission-powered explosion (atomic bomb trigger) produces intense X-rays that somehow are focused onto the fusion fuel (lithium deuteride, LiD). The X-rays presumably heat the fusion fuel and help compress it long enough for fusion reactions to occur. The technology for focusing the X-rays and other practical problems are so formidable that by knowing this piece of the Teller–Ulam "secret," one is only an infinitesimal distance along the way toward building a working superbomb.**

What converted Wheeler from a skeptic of black holes to a believer and advocate? Computer simulations of imploding stars were only the final validation of his conversion. Far more important was the destruction of a mental block. This mental block pervaded the world's community of theoretical physicists from the 1920s through the 1950s. It was fostered in part by the same *Schwarzschild singularity* that was then being used for a black hole. It was also fostered by the mysterious, seemingly paradoxical conclusion, from Oppenheimer and Snyder's idealized calculations, that an imploding star becomes frozen forever at the critical circumference ("Schwarzschild singularity") from the viewpoint of a static, external observer, but it implodes quickly through the freezing point and on inward from the viewpoint of an observer on the star's surface.

In Moscow, Landau and his colleagues, while believing Oppenheimer and Snyder's calculations, had severe trouble reconciling these two viewpoints. "You cannot appreciate how difficult it was for the human mind to understand how both viewpoints can be true simultaneously," Landau's closest friend, Evgeny Lifshitz, told me some years later.

Then one day in 1958, the same year as Wheeler was attacking Oppenheimer and Snyder's conclusions, there arrived in Moscow an issue of the *Physical Review* with an article by David Finkelstein, an assistant professor at a little known American university, the Stevens Institute of Technology in Hoboken, New Jersey. Landau and Lifshitzread the article. It was a revelation. Suddenly everything was clear.[5]

Finkelstein visited England that year and lectured at Kings College in London. Roger Penrose (who later would revolutionize our understanding of what goes on *inside* black holes; see Chapter 13) took the train down to London to hear Finkelstein's lecture, and returned to Cambridge enthusiastic.